CONTACT

Email: rahuljha190398@gmail.com

Phone: (782) 778-5607

Location: Calgary, AB (Open to Relocation)

LinkedIn: in/rahul-jha-480841168

Work Auth: Authorized to work in Canada

EDUCATION

M.Eng. Electrical & Computer

University of Calgary | 2024
GPA: 3.64 / 4.0 | Minor: Energy

B.Tech. Computer Science

REVA University | 2022
GPA: 8.70 / 10.0

LEADERSHIP & SYSTEMS

Technical Leadership:
Distributed Systems Design, Whiteboard Architecture, Technical Trade-Offs, Program Execution, Mentorship

Backend & Distributed:
Multi-threading, AsyncIO, Sockets (TCP/IP), Microservices, Modbus TCP, REST APIs, Rate Limiting

Languages & Core:
Python, C++, Go, Java, SQL, Bash, C#

Cloud & Infrastructure:
AWS, GCP, Docker, Kubernetes, CI/CD Pipelines, Linux/Unix, Distributed Tracing

Operational Excellence:
SLA Governance, High Availability (99.5%+), RCA, Health Daemons, Profiling

CERTIFICATIONS

- AWS Fundamentals (2020)
- Data Science Specialization (2020)
- Google Cloud 30-Day Track (2020)
- Blockchain Architecture (2020)

RAHUL JHA

Engineering Lead & Systems Architect | Distributed Backend, Program Delivery & Operational Scale

EXECUTIVE PROFILE

Hands-on Engineering Leader and Systems Architect holding a Master's degree in Electrical & Computer Engineering from the University of Calgary (GPA: 3.64). Combines deep backend technical roots with end-to-end ownership of complex engineering programs and high-availability distributed systems. Experienced in whiteboarding architectures, analyzing systemic trade-offs, diving deep into code-level bottlenecks, and delivering high-impact solutions across multi-threaded and cloud environments. Proven track record managing multi-stakeholder delivery, enforcing operational rigor, and maintaining 99.5%+ precision in high-scale production systems.

PROFESSIONAL EXPERIENCE

Lead Backend & Systems Developer (Technical Program Lead)

TR PRO | Calgary, AB | Jan 2023 – Present

- Owned technical architecture and delivery of a distributed execution engine utilizing multi-process Python and asynchronous scheduling, expanding processing throughput by 500% to handle peak loads.
- Engineered sub-millisecond execution logic, diving deep into thread safety, memory boundaries, and validation checks to prevent race conditions during real-time data ingestion.
- Drove program design and implementation for 8 dynamic webhook data stream engines, standardizing API contracts and data exchange across multi-threaded microservices.
- Built responsive client GUIs in Flutter paired with a robust Flask microservices backend, partnering with business stakeholders to reduce technical support overhead by 35%.

Project Engineer | Industrial Automation & Technical Programs

Litus Inc. | Calgary, AB | May 2024 – June 2025

- Led end-to-end program delivery of telemetry and automated data pipelines, processing 500,000+ monthly sensor data points via Modbus TCP protocols with zero data loss.
- Engineered multi-threaded Python automation daemons for real-time instrument telemetry, achieving 1,440 automated logs per unit daily and eliminating manual overhead.
- Analyzed multi-variable time-series process data (Python, Pandas, NumPy) to isolate system bottlenecks, driving a 30% optimization in system recovery efficiency.
- Designed specialized testing environments for analytical instrumentation (NMR equipment), creating robust REST APIs to synchronize edge analytics with central servers.
- Spearheaded custom internal ERP modules, diagnostics suites, and data visualization interfaces, streamlining daily cross-functional workflows by 40%.

Software Engineer | Financial Reliability & Transaction Systems

FIS | Bengaluru, India | Feb 2022 – Sept 2022

- Maintained 99.5% operational precision in automated end-of-day transaction pipelines for tier-one global financial institutions including RBC, BMO, and PayPal.
- Reduced system downtime by 20% by authoring proactive health-check diagnostics and automated fault-recovery scripts in Python and Bash.
- Configured mission-critical enterprise workflows in ZENA and established distributed Big Data connectors, accelerating issue resolution cycles by 15%.

Software Developer

Codertech | Bengaluru, India | Aug 2020 – Jan 2022

- Engineered cross-platform applications (Flutter/Dart) integrating secure REST APIs, scaling to support 60+ concurrent active video/data sessions.
- Integrated Cloud Firestore architecture and secure payment pipelines, boosting transaction success rates by 25% with end-to-end data validation.

Mobile Application Intern

REVA NEST Incubator | Bengaluru, India | Feb 2020 – May 2020

- Developed native Android modules using Java and Android SDK, optimizing local SQLite database caching and UI thread responsiveness.
- Implemented asynchronous background services for real-time JSON payload parsing and conducted multi-device QA profiling.